

1 Solution — GIAM 2.2

Problem 1

1.1 Problem

The transitive property of equality says that if $a = b$ and $b = c$ then $a = c$. Does the implication arrow satisfy a transitive property? If so, state it.

1.2 Answer

Yes. If $A \Rightarrow B$ and $B \Rightarrow C$, then $A \Rightarrow C$.

As a single tautology,

$$[(A \Rightarrow B) \wedge (B \Rightarrow C)] \Rightarrow (A \Rightarrow C).$$

This is a classical rule of inference called **hypothetical syllogism** (the “chain rule” for implications).

1.3 The Idea

Implication behaves just like $=$ here: it *chains*. If truth of A guarantees truth of B , and truth of B guarantees truth of C , then truth of A guarantees truth of C — passing through B as a stepping-stone.

Yes — implication chains. If A forces B and B forces C , then A forces C .

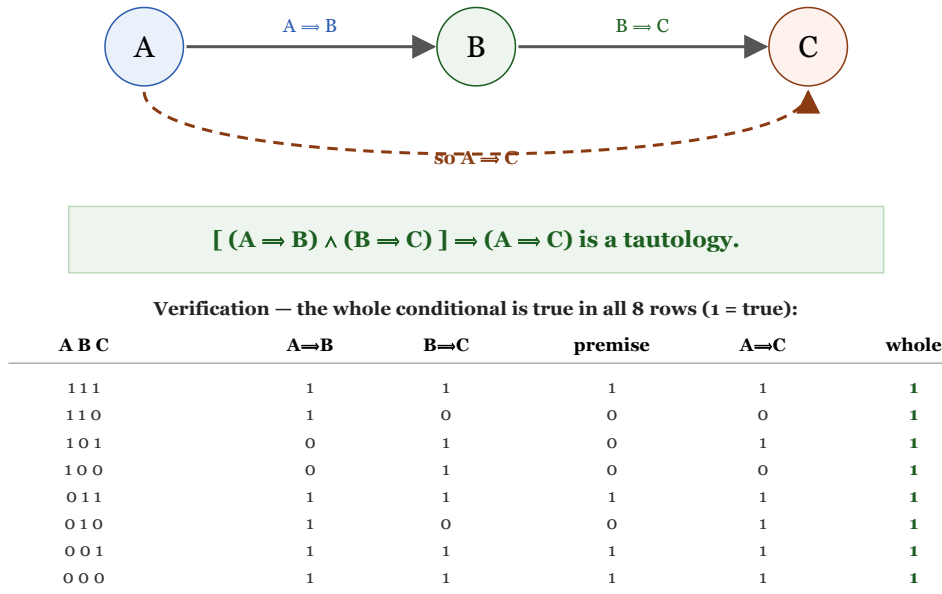


Figure 1.1: A chain diagram: A arrow B labelled A implies B , B arrow C labelled B implies C , and a dashed derived arrow from A to C labelled ‘so A implies C ’. Below, a boxed statement: $[(A \text{ implies } B) \text{ and } (B \text{ implies } C)] \text{ implies } (A \text{ implies } C)$ is a tautology. A truth table over all eight combinations of A, B, C lists the columns A implies B , B implies C , their conjunction (the premise), A implies C , and the whole conditional — whose final column is 1 in every one of the eight rows, confirming it is a tautology.

1.4 Verifying It Is a Tautology

Build the truth table for $[(A \Rightarrow B) \wedge (B \Rightarrow C)] \Rightarrow (A \Rightarrow C)$ over all 8 rows (recall $X \Rightarrow Y$ is false only when X is true and Y is false):

A	B	C	$A \Rightarrow B$	$B \Rightarrow C$	premise	$A \Rightarrow C$	whole
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	1
1	0	1	0	1	0	1	1
1	0	0	0	1	0	0	1
0	1	1	1	1	1	1	1

0	1	0	1	0	0	1	1
0	0	1	1	1	1	1	1
0	0	0	1	1	1	1	1

Table 1.1

The final column is 1 in **every** row, so the conditional is a tautology — the transitive property holds. ■

1.5 Remark — A Shorter Proof Than the Full Table

We need not check the outputs blindly; the conditional can only *fail* when its premise is true and its conclusion false. So suppose the premise $(A \Rightarrow B) \wedge (B \Rightarrow C)$ is true, and suppose (for contradiction) $A \Rightarrow C$ is false. " $A \Rightarrow C$ false" forces A true and C false. Then A true with $A \Rightarrow B$ true gives B true; and B true with $B \Rightarrow C$ true gives C true — contradicting C false. So the conclusion cannot fail while the premise holds: the implication is valid. This “attack only the one dangerous row” method is faster than an 8-row table and scales far better.

1.6 Remark — This Is Exactly How Proofs Are Built

Transitivity of $\Rightarrow$ is not a curiosity — it is the *engine* of essentially every mathematical proof. A proof is a chain

$$A \Rightarrow A_1 \Rightarrow A_2 \Rightarrow \cdots \Rightarrow A_n \Rightarrow Z,$$

and hypothetical syllogism, applied repeatedly, collapses the whole chain into the single conclusion $A \Rightarrow Z$. Each step licenses the next, and transitivity guarantees the end-points connect. When you prove “the hypotheses imply the theorem” by a sequence of intermediate lemmas, you are invoking this property at every link. The equality analogy in the problem is apt: just as $a = b = c = \cdots$ lets you conclude the first equals the last, $A \Rightarrow \cdots \Rightarrow Z$ lets you conclude the first implies the last.

1.7 Remark — Both $\Rightarrow$ and $\Leftrightarrow$ Are Transitive, but They Differ

The biconditional $\Leftrightarrow$ is *also* transitive — $(A \Leftrightarrow B) \wedge (B \Leftrightarrow C) \Rightarrow (A \Leftrightarrow C)$ — and it is the connective that most closely mirrors $=$, since it is additionally **symmetric** ($A \Leftrightarrow B$ gives $B \Leftrightarrow A$) and **reflexive** ($A \Leftrightarrow A$). Plain implication $\Rightarrow$ is transitive and reflexive ($A \Rightarrow A$) but **not** symmetric: $A \Rightarrow B$ does not give $B \Rightarrow A$ (rain implies clouds, but clouds do not imply rain). So $\Rightarrow$ behaves like an *order* relation (reflexive, transitive, antisymmetric up to equivalence) — akin to $\leq$ from Chapter 1 — while $\Leftrightarrow$ behaves like an *equivalence* relation, the true logical counterpart of equality.

1.8 Remark — The Contrast With a Non-Transitive Relation

It is worth appreciating that transitivity is a genuine, checkable property that some relations *lack* — so confirming it for $\Rightarrow$ is not automatic. “Is a friend of” is famously non-transitive (your friend’s friend need not be your friend); in logic, “is *consistent* with” fails too — A can be consistent with B and B consistent with C while A and C contradict. That $\Rightarrow$ *does* chain is precisely what makes long deductions trustworthy: you can stack a hundred valid steps and the conclusion still follows from the start. Had implication not been transitive, mathematics as a step-by-step enterprise would collapse.